

Four-Stock Quantitative Research Report

Adobe (ADBE) · Microsoft (MSFT) · Oracle (ORCL) · Salesforce (CRM)

1. Overview

This report presents an end-to-end quantitative research study of four technology stocks — Adobe, Microsoft, Oracle, and Salesforce — covering data validation, exploratory analysis, feature engineering, statistical research, machine learning, deep learning, backtesting, and Monte Carlo simulation. Each stock was analyzed independently using the same pipeline, then compared across all four to look for consistent or inconsistent patterns.

Stock	Data period	Rows (after cleaning)
Oracle (ORCL)	2019-01-02 to 2024-11-08	1,475
Salesforce (CRM)	2004-06-23 to 2024-11-29	5,146
Adobe (ADBE)	1986-08-13 to 2024-11-29	9,506
Microsoft (MSFT)	1986-03-13 to 2026-04-13	10,098

2. Methodology

For each stock, the same pipeline was applied:

- Load & audit** — read the raw CSV, check for missing values, duplicate rows/dates, and invalid OHLC relationships (e.g. High below Close).
- Clean** — remove any rows that failed the above checks.
- Descriptive statistics** — CAGR, annualized volatility, maximum drawdown, skew, and kurtosis of daily returns.
- Exploratory analysis (Seaborn)** — price trend, return distribution, correlation heatmap, and volume trend.
- Feature engineering** — moving averages, momentum, rolling volatility, RSI, MACD, Bollinger Bands, ATR, gap, and volume ratio.
- Statistical research** — weekday/month return patterns, and Pearson correlation of each engineered feature against next-day return, with p-values.
- Machine learning** — Logistic Regression and Random Forest for next-day direction (up/down); Linear Regression and Random Forest for next-day return value. All models were validated on a held-out final 20% of the timeline (no shuffling, since this is time-series data) and compared against a naive baseline (always predict the majority class / average return).

8. **Regime & anomaly detection** — PCA + K-Means to group days into market regimes; Isolation Forest to flag statistically unusual days.

9. **Backtesting** — converted the Random Forest's predicted probability into a simple long/cash signal, then compared its Sharpe ratio, Sortino ratio, CAGR, and max drawdown against passive Buy & Hold over the same test period.

10. **Monte Carlo simulation** — simulated 10,000 possible one-year-ahead price paths using the stock's own historical mean and volatility.

11. **Deep learning** — a small Dense network and an LSTM, trained on the same engineered features, compared against the classical ML baselines on RMSE.

3. Per-Stock Summary

Metric	ORCL	CRM	ADBE	MSFT
CAGR	27.72%	23.69%	18.33%	22.92%
Annualized volatility	31.03%	41.93%	48.23%	33.21%
Max drawdown	-41.08%	-70.50%	-79.89%	-74.56%
Positive days	53.56%	51.30%	49.85%	50.53%
Return skew	1.114	0.348	0.141	-0.103
Return kurtosis	18.72	10.14	10.05	10.83
Anomaly days flagged	30	103	190	202

Oracle stands out with the lowest volatility and drawdown of the four, largely because its dataset only covers 2019 onward and therefore excludes the dot-com crash and 2008 financial crisis that appear in the longer ADBE, CRM, and MSFT histories. Adobe's -79.9% drawdown and Microsoft's -74.6% both reflect those earlier crashes. All four stocks show positive skew or near-neutral skew and high kurtosis, meaning large single-day moves are more common than a normal distribution would predict — a well-known feature of real stock data, not a data quality issue.

4. Machine Learning Results

Metric	ORCL	CRM	ADBE	MSFT
Logistic Regression accuracy	48.45%	51.90%	53.61%	47.82%
Random Forest accuracy	48.80%	49.56%	51.08%	51.88%
Baseline accuracy (majority class)	56.01%	51.61%	46.55%	46.68%
Linear Regression RMSE	0.02099	0.02295	0.02360	0.02086
Random Forest RMSE	0.02320	0.02471	0.02722	0.02362
Baseline RMSE (predict average)	0.02054	0.02268	0.02241	0.01786
Linear Regression R-squared	-0.046	-0.025	-0.109	-0.365
Random Forest R-squared	-0.280	-0.189	-0.475	-0.749

No model beat its baseline on every stock. Adobe and Microsoft were the two cases where Random Forest classification beat the majority-class baseline (51.1% vs 46.6%, and 51.9% vs 46.7%); Salesforce's Logistic Regression narrowly beat its baseline (51.9% vs 51.6%); Oracle's classifiers underperformed baseline on all counts. Every regression model's R-squared was negative across all four stocks — meaning none of the models explained next-day return variance better than simply predicting the historical average return every day.

5. Deep Learning Results

Metric	ORCL	CRM	ADBE	MSFT
Dense network RMSE	0.02248	0.02429	0.02749	0.01836
LSTM RMSE	0.02156	0.02314	0.02338	0.01796
Baseline RMSE	0.02054	0.02268	0.02241	0.01786

Neither the Dense network nor the LSTM beat the naive baseline on any of the four stocks. Both deep learning models also showed run-to-run variance of a few thousandths in RMSE due to unseeded weight initialization — a known TensorFlow characteristic, not a bug — but the conclusion (deep learning underperforming the baseline) held consistently across repeated runs.

6. Backtesting Results

Metric	ORCL	CRM	ADBE	MSFT
Strategy Sharpe	2.133	0.513	0.584	0.384
Buy & Hold Sharpe	1.515	0.424	0.681	0.766
Strategy CAGR (test period)	37.10%	9.84%	12.33%	4.36%
Buy & Hold CAGR (test period)	55.51%	9.12%	19.57%	19.37%
Strategy max drawdown	-6.74%	-31.26%	-36.18%	-29.14%
Buy & Hold max drawdown	-14.38%	-58.62%	-60.02%	-37.56%

The model-driven strategy beat Buy & Hold on risk-adjusted return (Sharpe ratio) for Oracle and Salesforce, in both cases with a noticeably smaller drawdown than passive holding. For Adobe and Microsoft, Buy & Hold won on both raw return and Sharpe ratio — both stocks had a long, steady multi-decade climb in their test windows that a cautious, signal-gated strategy could not fully capture. This is a genuinely mixed result: the strategy's main advantage across all four stocks was **risk reduction**, not higher returns, and that advantage was not consistent enough to claim the model reliably beats passive investing.

7. Monte Carlo Simulation (One Year Ahead)

Metric	ORCL	CRM	ADBE	MSFT
Starting price	\$189.25	\$329.99	\$515.93	\$380.52
Mean simulated ending price	\$253.92	\$446.86	\$688.31	\$495.41
5th percentile (bad case)	\$144.45	\$203.82	\$274.79	\$269.82
95th percentile (good case)	\$402.42	\$814.51	\$1,352.13	\$808.29
Probability price ends higher	78.94%	69.84%	63.76%	73.82%

These simulations project forward using each stock's own historical mean return and volatility, assuming that pattern continues — they are not predictions, just a way of visualizing the range of outcomes implied by each stock's historical risk/return profile.

8. Central Finding

Across all four stocks, **next-day return prediction remained weak out of sample**. Classification accuracy beat a naïve majority-class baseline in only some cases, and every regression model — classical and deep learning alike — produced a negative R-squared, meaning none of them explained next-day return variance better than simply predicting the historical average. Deep learning (Dense and LSTM) did not consistently outperform simpler classical models; in several cases it performed worse. The backtested trading strategy reduced risk (smaller drawdowns, and in two of four cases a better Sharpe ratio) but did not reliably beat Buy & Hold on raw returns.

This is reported as the primary finding rather than something to work around: short-horizon stock return prediction is genuinely difficult, and these results are consistent with the broader body of research on market efficiency. The value of this study is in the honest, reproducible measurement of that difficulty across four different companies, data lengths, and volatility regimes — not in producing a model that reliably beats the market.

9. Limitations

- No transaction costs, slippage, or taxes were modeled in the backtest beyond the basic long/cash signal structure.
- The Monte Carlo simulation assumes returns are i.i.d. and normally distributed, which understates the fat-tailed, clustered-volatility behavior actually observed in the return distributions (see Section 3).
- Feature set and model architecture were kept identical across all four stocks for comparability; stock-specific tuning was intentionally not performed.
- Datasets differ substantially in length (6 years for Oracle vs. ~38–40 years for Adobe and Microsoft), which affects how much market history (crashes, recoveries) each analysis captures.